

Subject Description Form

Subject Code	ITC3222K
Subject Title	Knitted Structure Design
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Co-requisite: ITC3226K Knitwear Technology I and ITC3227K Knitwear Technology II Exclusion: SFT316KD Knitted Structure Design
Objectives	The subject provides the foundation knowledge on knitting techniques and fabrics through hands-on exercises on V-bed hand knitting machine, basic Circular and Warp knitting. It covers the essential elements of the formation, properties and end-uses of knitted fabrics for the production of knitwear.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) describe and elaborate the basic principles of knitted fabric formation; (b) define and explain the essential terms in weft/warp knitting fabrics/panels manufacture; (c) identify and compare the different knitted fabric structures and properties; (d) communicate effectively with industrial practitioners; (e) develop a foundation for further study and engage in life-long learning; and (f) identify and apply the safety and ethical practices in the industry.
Subject Synopsis/ Indicative Syllabus	(I) V-bed Hand Knitting Machine and Suitable Yarn for Knitwear Specification, configuration and operation of V-bed hand knitting machines. Overview of the yarn types suitable for the non-jacquard fabrics and the relationship between yarn type, fabric property and fabric application. Basic and complicated flat knitting fabric structures, including ribs, plain double jersey and single jersey, lace, cable, racking, jacquard, terry; fringe. Partial knitting, plating knitting and fully-fashioned knitting.

	(II) Single jersey circular Knitting Machine and Suitable Yarn for Knitwear Specification, configuration and operation of single jersey circular knitting machine Overview of the yarn types suitable for the non-jacquard fabrics and the relationship between yarn type, fabric property and fabric application. (III) Tricot warp Knitting Machine and Suitable Yarn for Knitwear Specification, configuration and operation of 3-bar tricot warp knitting machine Overview of the yarn types suitable for the non-jacquard fabrics and the relationship between yarn type, fabric property and fabric application. (IV) Developing Skills for Knitted Structure Mix and combination of yarn, fabric structure and knitting technique. Creative fabric structure design project. (V) Safety and Ethical Practices in the Industry General safety and general ethical practices when operating the machines in knitwear industry
Teaching/Learning Methodology	Laboratory work will be carried out in the knitting workshop with demonstration and guidance from lecturer/technician. Students will work on a project base and will need to submit their designed fabrics at the end for the final assessment.

**Reading List and
References**

Books

Brackenbury, T. (1992), *Knitted Clothing Technology*. Clackwell Scientific Publications, Oxford.

Lee, R. (2003), *Creative machine knitting*. Guild of Master Craftsman, Lewes England.

Raz, S. (1993), *Flat knitting technology*. Universal Maschienenfabrik, Westhausen.

Spencer, D. J. (2001), *Knitting technology a comprehensive handbook and practical guide*. 3rd Edition, Woodhead publishing Ltd.

Yue, K. H. (1995), *Knitting Handbooks No.7, No. 9 & 10*. The Hong Kong Polytechnic University.